

Hot metal

Ignite a mixture of chemicals called thermite and you'll need to stand back! These chemicals react together very quickly, producing enormous amounts of heat.

A thermite reaction is a spectacular display, but also serves a practical purpose: it is used to extract molten iron from iron oxide for welding. It takes a lot of heat to start the reaction, which then releases enough heat to melt the iron. The process most commonly uses a mixture of iron oxide and aluminum. A slim ribbon of magnesium is inserted into the mixture as a fuse. When ignited, it starts the reaction, breaking the bonds between the iron and oxygen atoms. Aluminum then bonds with the released oxygen, producing more heat. This in turn breaks more bonds and melts the leftover iron.

